The system-level criteria cover three aspects of operation. Cut-height and lateral errors measure task accuracy. Contact, head alignment, impact rate, and cutter acceleration describe tool–terrain interaction and motion quality. Cylinder headroom and safety terminations check whether the task is completed without driving the mechanism into its limits. The metrics are evaluated over the scored part of each episode and are kept in physical units where possible. This makes a failed gate traceable to a specific limitation rather than to a single combined reward value.

3.9.1 Eight-Gate Acceptance Contract

Table 3.5 defines the system-level acceptance contract.

All controller stacks are evaluated with the same path, terrain, and randomization conditions. A stack passes only when it satisfies every gate; strong mean tracking cannot compensate for loss of contact, inadequate cylinder headroom, excessive impacts, or safety terminations. Worst-case aggregation prevents easy terrain or path cases from masking a local failure. Mean absolute errors quantify sustained tracking performance, while the lower and upper percentiles capture short alignment, clearance, or acceleration events that a mean can hide. Contact and termination measures are reported as fractions so that episodes of different completed length remain comparable.

Table 3.5: Eight system-level acceptance gates. Each gate uses at least three terrain meshes and the worst mesh; tracking-critical gates also use the worst path family. †Hard impacts is a wear constraint.

Gate	Metric	Threshold	Aggregation
Cut height	mean $\lvert e_{\mathrm{cut}} \rvert$	$\leq 0.040\,\mathrm{m}$	worst mesh
Lateral tracking	mean $\lvert e_{\mathrm{lat}} \rvert$	$\leq 0.300\,\mathrm{m}$	worst mesh, worst family
Ground contact	contact fraction	≥ 0.700	worst mesh, worst family
Head alignment	p01 head-down alignment	≥ 0.900	worst mesh
Hard impacts†	hard-impact step fraction	≤ 0.350	worst mesh
Tool acceleration	p95 cutter acceleration	$\leq 120\,\mathrm{m/s^2}$	worst mesh
Cylinder headroom	p01 travel margin	$\geq 0.005\,\mathrm{m}$	worst mesh
Safety terminations	termination rate	≤ 0.080	worst mesh

3.9.2 Cylinder and Model Checks

Cylinder checks use fresh $\pm 10\,\mathrm{mm}$ commands, coordinated $80\,\mathrm{mm}$ motions, return error and saturation. Model checks use one-step and rollout accuracy, input ablation, and Jacobian conditioning (Chapter 5). Chapter 7 reports these with the acceptance contract. The small command checks local response and deadband; the coordinated motion checks coupling between actuator channels. Return error tests reversibility, and saturation checks whether the learned model respects the command and stroke limits. These diagnostics are interpreted alongside the system gates, not as substitutes for closed-loop task performance.

The system is now specified. The next chapter turns to the data the learned models are built from.